

SCIENTIFIC SEARCH | SYNTHESIS | EMPIRICAL DISCIPLINE

CROSS-SCIENCE MARKET REGIME DISCOVERY WITH A 50-AGENT SWARM

Searching Across Scientific Domains for a Novel, Falsifiable Specification of Latent Financial Market Regimes

A pedagogical research paper for financial professionals on how artificial-intelligence swarms can expand the scientific hypothesis space, synthesize cross-disciplinary mechanisms, and submit the resulting architecture to empirical validation.

RESEARCH OBJECTIVE

This paper investigates whether a structured swarm of specialized artificial-intelligence agents can improve the design of a financial market-regime model by searching beyond the conventional finance toolkit. Fifty agents, organized into ten scientific families, are asked to reinterpret latent-state detection through the mature mathematical traditions of statistical physics, dynamical systems, signal engineering, control theory, information theory, applied mathematics, chemistry, astrophysics, network science, and geophysics. Their proposals are not averaged mechanically. A separate synthesis stage identifies recurring mechanisms, complementary components, minority ideas worth preserving, and falsifiable empirical implications. The resulting **Metastable Basin-Transition Architecture (MBTA)** is then implemented as an ensemble of dimensionality reduction, probabilistic and geometric state inference, transition-pressure measurement, and explicit model-disagreement analysis. The objective is not to proclaim a final ontology of markets, but to demonstrate a disciplined method for moving from broad scientific exploration to a novel, executable, and challengeable regime specification.

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Abstract

Market regimes are latent: investors observe returns, volatility, spreads, correlations, breadth, and cross-asset rotations, but not the underlying state itself. Conventional financial research therefore begins with a model class—for example, a Hidden Markov Model, a Markov-switching process, a mixture model, or a change-point detector—and asks whether that model produces useful states. This paper reverses the order of inquiry. It first asks how other scientific disciplines conceptualize hidden states and transitions, then uses a fifty-agent artificial-intelligence swarm to explore that broader hypothesis space. Ten disciplinary families contribute five specialized perspectives each. Every agent produces a structured proposal specifying the latent object, transfer principle, mathematical core, required financial observables, transition mechanism, expected state count, validation tests, failure modes, novelty, plausibility, and confidence. A separate synthesis agent then searches for cross-disciplinary recurrence and complementarity rather than selecting a winner by majority vote. The resulting Metastable Basin-Transition Architecture combines low-dimensional order-parameter representation, Gaussian-mixture, K-means and spectral state inference, explicit transition intensity, and model disagreement as an uncertainty signal. Applied to 3,641 observations and thirteen standardized market features, the experiment identifies four persistent latent states. Consensus persistence is 0.873; mean perturbation Adjusted Rand Index is 0.870; the GMM silhouette is 0.176; model disagreement is 2.141 times higher on switch observations; and the ranges of mean 21-day forward log return and forward realized volatility across regimes are 0.0297 and 0.1487, respectively. The results are encouraging but deliberately mixed: the architecture is persistent and generally robust, yet geometric separation is modest and local perturbation robustness is not universal. The paper’s principal contribution is therefore methodological. AI is used to expand scientific imagination; deterministic code and falsification-oriented diagnostics are used to constrain it.¹

1. Introduction: Regime Detection as a Scientific Search Problem

Financial professionals routinely reason in regimes. Risk budgets change when volatility rises; diversification assumptions are reconsidered when correlations converge; credit, equity, rates, commodities, and currencies acquire different meanings when the macro-financial environment changes. Yet the variable that practitioners care about most—the underlying market condition—is not directly observed. What is observed is a stream of imperfect manifestations: returns, drawdowns, realized volatility, implied volatility, credit behavior, breadth, cross-asset dispersion, correlation, and relative rotations among defensive and risky assets.

This makes regime detection a latent-state inference problem. Let $x_t \in \mathbb{R}^p$ denote the vector of observable market features at time t and let z_t denote an unobserved state. The research problem is not simply to cluster x_t . It is to infer whether there exists a useful representation

$$z_t = \mathcal{G}(x_t, x_{t-1}, \dots; \Theta)$$

whose states are sufficiently persistent, interpretable, stable, and economically differentiated to deserve the word *regime*. The mapping $\mathcal{G}$ is unknown; so is the appropriate geometry of the latent space, the number of states, the transition mechanism, and even the correct ontology of what a regime should mean.

The conventional response is model selection. One might begin with a Hidden Markov Model, estimate two to five states, compare information criteria, and interpret the resulting transition matrix. This is legitimate and important. Hamilton’s Markov-switching framework and the broader HMM literature provide foundational tools for hidden-state inference. But beginning with a model class silently narrows the scientific question. It assumes, before the investigation begins, that the most useful latent structure resembles the chosen model.

The research objective of this paper is different: **to explore a deliberately broad universe of scientific domains in search of an idoneous specification for a novel market-regime detection architecture**. The word *idoneous* is used carefully. We do not mean universally optimal. We mean scientifically defensible for the stated problem: broad enough to capture distinct mechanisms, parsimonious enough to implement, interpretable enough to audit, and explicit enough to falsify.

The notebook operationalizes this objective with a fifty-agent swarm. The swarm is not used because fifty opinions are automatically superior to one. It is used because the agents are assigned genuinely different intellectual priors. Statistical physicists are asked to think in phases, order parameters, metastability, and critical phenomena. Dynamical-systems specialists think in attractors, bifurcations, and slow-fast systems. Signal engineers think in spectra, wavelets, source separation, and change detection. Control theorists think in observability, state-space systems, filtering, and system identification. Information theorists think in entropy, divergence, transfer entropy, and compression. Applied mathematicians contribute operator methods, topology, optimal transport, manifolds, and inverse problems. Chemistry contributes reaction coordinates, transition states, free-energy landscapes, and nucleation. Astrophysics contributes weak-signal extraction and multiscale latent-source inference. Network science contributes correlation topology, percolation, communities, and dynamic graphs. Geophysics contributes precursor detection, stress accumulation, rupture, and regime shifts in complex systems.

The swarm therefore changes the first question from “Which financial model should we fit?” to “What kinds of

¹The companion Google Colab notebook is available at <https://drive.google.com/file/d/1JoLlJtV0ufUdlyNfAfiTosSW8CWvwuAs/view?usp=sharing>.

hidden-state mechanisms have mature sciences already learned to recognize?” The remainder of the paper explains how that expansion is organized, how the fifty proposals are compressed into one architecture without destroying minority insights, how the architecture is translated into deterministic empirical code, and what the resulting performance tells us.

Several criteria are needed before an inferred state deserves to be treated as a regime rather than as a convenient cluster label. First, the state should exhibit temporal persistence. A partition that changes almost every day may describe local geometry, but it does not correspond to the slower-moving environments that motivate strategic risk decisions. Second, the state should possess multivariate coherence. A regime should be visible through a configuration of indicators rather than through one arbitrary threshold. Third, it should have economic differentiation: the distributions of risk, correlation, liquidity, or subsequent outcomes should differ in ways that matter for interpretation. Fourth, it should be reasonably stable to small perturbations of the data and modeling choices. Finally, the state should remain uncertain where the evidence is ambiguous. A credible system must be capable of saying that several interpretations are plausible.

These requirements expose why regime research is scientifically difficult. Persistence can be manufactured through excessive smoothing. Separation can be improved by adding more clusters until each observation occupies a narrow region. Economic differentiation can be exaggerated through in-sample selection. Interpretability can be imposed after estimation by attaching attractive names to anonymous labels. Robustness can appear strong when only one kind of perturbation is tested. Every desirable property can therefore be obtained superficially. The challenge is to construct a collection of diagnostics whose weaknesses constrain one another. Persistence without differentiation is not enough; differentiation without stability is not enough; and stability without out-of-sample validity is not enough.

The latent-state perspective also changes the role of uncertainty. In a supervised problem, prediction error can be evaluated against observed labels. In regime detection, the label is a model output. Two algorithms can disagree because one is wrong, because both are incomplete, or because the observation truly lies near a boundary at which categorical language becomes strained. For that reason, the research does not force all uncertainty into a posterior probability from one model. It preserves disagreement across models with different geometries. This converts uncertainty from an inconvenient residue into an observable feature of the regime system.

The cross-science strategy is particularly useful because scientific disciplines encode different answers to four questions: what constitutes a state, what keeps a system in that state, what causes a transition, and what evidence signals that a transition is approaching. Statistical physics emphasizes collective variables and metastability. Dynamical systems emphasizes attractors, basins, and bifurcations. Signal processing emphasizes decomposition and detection under noise. Control theory emphasizes observability and state estimation. Information theory emphasizes distinguishability between distributions. Network science emphasizes changes in relational structure. None of these traditions can be transferred mechanically to markets, but each supplies disciplined concepts that can generate testable financial analogues.

This approach does not reject established financial models. Hidden Markov Models, Markov-switching regressions, change-point methods, mixture models, and threshold rules remain essential benchmarks. The contribution is to place model selection inside a larger design process. A conventional model can appear as one candidate implementation of a scientific mechanism rather than as the unquestioned starting ontology. This distinction matters because two models with similar statistical performance may imply very different ideas about transition risk, uncertainty, and intervention.

The intended standard is therefore institutional rather than merely algorithmic. A useful research architecture should document where ideas came from, which alternatives were rejected, how labels were formed, which diagnostics challenged them, and what future evidence would justify revision. The swarm is one mechanism for creating this documented plurality. Its success should be judged not by the rhetorical novelty of its output but by whether it produces a more explicit, auditable, and falsifiable path from conjecture to evidence.

The methodological principle is expansion before compression. Artificial intelligence is used first to enlarge the scientific hypothesis space. Mathematical structure, common data, executable constraints, and empirical falsification then compress that space into a research architecture that can survive or fail in contact with evidence.

2. Empirical Substrate: Giving Fifty Scientists the Same Market

Notebook mapping: Cell 1 establishes the environment and acquires the data; Cell 2 engineers and standardizes the common feature set.

A cross-disciplinary search becomes meaningless if every agent imagines a different empirical world. The notebook therefore establishes a common data substrate before scientific ideation begins. The investable and informational universe spans equities, rates, credit, commodities, the dollar, gold, and implied volatility. Representative instruments include SPY, QQQ, IWM, TLT, IEF, LQD, HYG, GLD, DBC, UUP, and the VIX.

The purpose is not to create a maximal market database. It is to create a compact measurement system rich enough to represent several economically distinct dimensions of state. Thirteen standardized features are constructed: five-day equity return, sixty-three-day equity trend, twenty-one-day realized volatility, downside volatility, drawdown, cross-asset dispersion, average correlation, breadth, VIX level, VIX change, credit risk, bond-versus-equity rotation, and gold-versus-equity rotation. These features deliberately mix level, momentum, dispersion, dependence, risk, and defensive-rotation information.

For a financial professional, the distinction is important. A regime is not synonymous with whether equities rose yesterday. A high-volatility deleveraging state may combine negative trend, deep drawdown, rising correlation, deteriorating breadth, elevated VIX, and defensive cross-asset rotation. Conversely, a constructive state may combine positive trend, broad participation, lower volatility, and contained correlation. The object being inferred is therefore a configuration of the market system.

Let F_t denote the raw feature vector. The notebook applies robust scaling to reduce sensitivity to extreme observations. Conceptually,

$$X_{tj} = \frac{F_{tj} - \text{median}(F_j)}{\text{IQR}(F_j)},$$

where X_t is the standardized state vector. Robust scaling is especially appropriate for financial data because crisis observations are informative but can dominate mean-and-standard-deviation normalization.

This separation between theory and measurement is central to the design. The agents are free to propose different latent mechanisms, but they must explain how those mechanisms connect to the same observable features. Scientific creativity is therefore constrained by a common empirical contract.

The feature system is designed around distinct measurement roles. Return and trend features represent direction and persistence in risky assets. Realized volatility and downside volatility represent the magnitude and asymmetry of fluctuations. Drawdown records path dependence that a contemporaneous return cannot capture. Dispersion and average correlation describe whether assets are behaving idiosyncratically or moving as a tightly coupled system. Breadth measures whether an equity move is broadly shared. VIX variables provide an options-market view of anticipated risk. Credit and defensive-rotation measures connect equity conditions to the behavior of bonds, gold, commodities, and the dollar. Taken together, these variables form a compact state vector rather than a collection of interchangeable technical indicators.

The common substrate serves an experimental-control function. If one agent were allowed to use only price trends while another imagined dealer gamma, funding spreads, and intraday order books, their proposals could not be compared fairly. Their apparent scientific differences would be confounded with different information sets. By holding the observable system fixed, the experiment asks a cleaner question: given the same evidence, how do different scientific traditions define the hidden object and its transitions? Later research can broaden the evidence set, but the first comparison requires this discipline.

Timing is equally important. Every feature must be computable with information available at the observation date. Rolling statistics use trailing windows, drawdown depends only on the historical price path, and cross-asset measures are synchronized before estimation. Missing values created by different trading calendars or lookback requirements must be handled before the matrix is presented to the models. These details are not merely engineering. A subtle forward fill across an inappropriate gap, an unsynchronized close, or a centered rolling window can create information that would not have existed in real time and thereby corrupt the interpretation of regimes.

Robust scaling is used because the empirical distribution of market variables is neither Gaussian nor stationary. A crisis observation should remain extreme, but it should not determine the units in which every other observation is measured. Median and interquartile-range normalization reduce the leverage of isolated extremes while retaining their directional meaning. The transformation is applied feature by feature, so it does not erase dependence across variables. In a production walk-forward study, the scaler would have to be fitted only on the training sample and then frozen or updated under a declared rule. The present full-sample implementation is appropriate for descriptive architecture testing but not for a final investable claim.

The feature matrix also embodies an unavoidable modeling choice about horizon. Five-day return, twenty-one-day volatility, and sixty-three-day trend do not evolve at the same speed. This mixture is intentional because a market state can contain fast stress and slow trend simultaneously. Yet it means that the inferred regimes are not tied to one unique clock. An alternative design could construct separate fast, medium, and slow latent layers. The compact implementation instead lets PCA and the clustering models discover combinations across horizons. This improves parsimony at the cost of potentially blending processes that would be separated in a hierarchical model.

Data quality must be understood as part of model risk. Exchange-traded funds can change liquidity, composition, or inception coverage. VIX and cash-market instruments have different microstructures. Total-return adjustments, holiday alignment, stale prices, and symbol changes can affect derived signals. A reproducible pipeline therefore needs a data dictionary, source timestamps, transformation definitions, and diagnostic reports for missingness and outliers. The notebook's environment and acquisition cell is mapped explicitly because the scientific chain begins here: an elegant latent-state theory cannot rescue an incoherent measurement system.

Finally, the substrate is deliberately modest. Thirteen features are sufficient to demonstrate cross-disciplinary

synthesis without creating a high-dimensional search problem in which almost any partition can be rationalized. This restraint supports interpretability and makes sensitivity analysis feasible. The cost is omitted-variable risk. Funding conditions, volatility surfaces, macroeconomic surprises, liquidity measures, positioning, and international markets may contain state information absent from the prototype. The correct interpretation is not that the thirteen variables exhaust the market, but that they define the controlled observational language for this experiment.

A formal data freeze would strengthen the design further. The research record should preserve the retrieval date, raw files, adjustment policy, trading calendar, transformation code, and final observation set. Revisions to any input should create a new dataset version rather than silently altering the historical experiment. This practice would let future researchers distinguish changes caused by data corrections from changes caused by model development, and it would make the common empirical contract genuinely reproducible across reruns.

3. The 50-Agent Swarm: Systematic Exploration of Scientific Domains

Notebook mapping: Cell 3 defines the swarm, Cell 4 tests one agent and validates the response contract, and Cell 5 launches and checkpoints the full swarm.

3.1 Ten families, five specialists per family

The swarm contains fifty agents distributed across ten scientific families. Within each family, five specialists are assigned distinct subdomains. Statistical physics, for example, is decomposed into phase transitions and order parameters; metastability and energy landscapes; critical phenomena and susceptibility; spin systems and collective alignment; and non-equilibrium statistical mechanics. Dynamical systems contributes attractors and basins, bifurcation theory, state-space reconstruction, Lyapunov stability, and slow-fast systems. Similar specialization is applied to the remaining families.

This design prevents a common failure mode in multi-agent systems: nominal diversity without intellectual diversity. Fifty agents receiving the same generic prompt would create repeated samples from approximately the same distribution. Here, diversity is architectural. The role definition changes the prior from which each agent reasons.

3.2 Structured scientific proposals

Every agent must return the same research schema. The required fields include the proposed model name, the latent object, the scientific transfer principle, the mathematical core, required financial features, a transition detector, an expected state count, validation tests, failure modes, and self-assessments of novelty, plausibility, and confidence. The result is not fifty essays. It is a structured corpus of fifty scientific hypotheses.

This is more important than it may initially appear. A swarm becomes scientifically useful only when its outputs can be compared. Structured output makes the proposal population analyzable: one can count preferred state numbers, inspect recurring mathematical vocabulary, compare validation logic, and preserve provenance from agent identity to final synthesis.

The notebook uses Claude Sonnet 5 through a structured JSON response contract. A one-agent canary is executed before the complete swarm. This small engineering step has methodological significance: it prevents an interface or schema error from being multiplied across fifty expensive calls. Each successful proposal is checkpointed, so interruption does not erase completed scientific work.

3.3 Breadth rather than maximal depth per agent

Individual agents are run with low reasoning effort and concise output requirements. This is intentional. The experiment allocates intelligence to population diversity rather than to a single extremely long chain of reasoning. The design resembles portfolio diversification: the objective is not to maximize the conviction of one forecaster, but to sample materially different sources of scientific structure.

The fifty-agent launch therefore creates an explicit hypothesis-generation stage. The output of Cell 5 is a scientific population, not a model. At this point the notebook has intentionally increased uncertainty: instead of one regime specification, it has fifty.

The swarm's constitution is designed before any agent is called. Each family receives a distinct scientific identity, and each specialist receives a narrower mechanism within that family. This matters because diversity cannot be evaluated only after the outputs arrive. If the prompts are nearly identical, apparent agreement may simply reflect a shared prompt template or the language model's default associations. By declaring the population in advance, the notebook creates a sampling frame for ideas. The fifty responses can then be interpreted as conditioned draws from an intentionally heterogeneous set of intellectual priors.

The structured schema performs several roles at once. It forces every proposal to name its latent object, preventing metaphor from remaining vague. It requires a transfer principle, so the agent must explain why a concept from another science has a meaningful financial analogue. It requests a mathematical core and required observables, connecting verbal reasoning to implementable quantities. It asks for a transition detector, validation tests, and failure

modes, preventing the proposal from focusing only on state classification. Finally, numerical self-assessments provide metadata that can be compared with the substance of the response, although they are not treated as calibrated probabilities.

The one-agent canary separates interface validation from scientific evaluation. Before launching fifty calls, the notebook verifies that the selected agent identity is preserved, required arrays are populated, scores lie within their allowed ranges, the expected state count is plausible, and every requested market feature exists in the empirical system. A failure at this stage is repaired in the prompt, schema, or API interface. This avoids contaminating the scientific corpus with structurally invalid responses and makes cost control part of methodological rigor.

Checkpointing gives the swarm an operational memory. Each accepted proposal is written to a durable artifact together with the agent identifier and family. If execution stops, completed proposals can be recovered without silently regenerating them. This is important because language-model outputs are stochastic even under similar settings. Re-running a missing subset is not equivalent to re-running the whole population, and provenance must record which calls produced the corpus actually synthesized. The research object is therefore not just a table of ideas; it is a traceable set of calls, validations, and saved outputs.

Low reasoning effort per agent is a deliberate allocation choice, but it creates a tradeoff. Broad sampling increases conceptual coverage, whereas deep deliberation may improve the internal quality of an individual proposal. The experiment chooses breadth because its purpose is to search for recurring mechanisms across domains. A later stage could allocate additional computation to proposals that survive initial screening, much as a scientific organization moves from exploratory grants to concentrated research programs. The current swarm should therefore be understood as a first-pass map of the hypothesis landscape rather than fifty completed theories.

Several forms of dependence remain possible. The agents share a base language model, common instructions, the same feature list, and broad cultural knowledge. Their errors are not independent in the statistical sense. A phrase that is strongly associated with “regime change” in the training distribution may recur across families even when the underlying scientific traditions differ. The synthesizer must therefore distinguish meaningful cross-domain convergence from lexical repetition. Recurrence becomes more credible when the same structural role appears through different terminology and when the proposed mechanism leads to distinct observable tests.

The population also preserves minority proposals. A mechanism suggested by only one or two agents can be valuable if it addresses a layer neglected by the majority, such as uncertainty, causal direction, topology, or transition propagation. The swarm is not a referendum. Its purpose is to make the design space visible enough that a subsequent synthesis can compare recurrence, complementarity, implementability, and falsifiability. This is why the full output remains part of the research provenance even though only a subset of its ideas enters MBTA.

4. From Fifty Scientific Ideas to One Research Architecture

Notebook mapping: Cell 6 synthesizes the fifty proposals into one falsifiable, executable architecture.

4.1 Why majority voting would be the wrong synthesis rule

The simplest aggregation rule would be to count proposals and select the most popular idea. That would undermine the purpose of the experiment. Scientific progress is not guaranteed to be majoritarian. A minority proposal based on persistent homology, transfer entropy, or nucleation could contribute a mechanism absent from the modal response. The synthesis must therefore distinguish recurrence from conformity.

Cell 6 introduces a separate scientific synthesizer. Its mandate is architectural: identify recurring structures, complementary mechanisms, minority ideas worth preserving, and a parsimonious empirical implementation. The synthesizer is constrained to a finite set of executable components—PCA, Gaussian mixtures, K-means, spectral clustering, transition intensity, and consensus voting—so that scientific metaphor cannot escape empirical implementation.

The synthesis is judged along three dimensions. First, **cross-disciplinary recurrence**: ideas that emerge independently in unrelated sciences may reveal structural features of latent-state problems. Second, **complementarity**: representation, classification, transition detection, and uncertainty need not be solved by the same mechanism. Third, **falsifiability and implementability**: an elegant analogy is insufficient unless it generates observable implications and can be rejected.

4.2 The Metastable Basin-Transition Architecture

The synthesis produced the **Metastable Basin-Transition Architecture (MBTA)**. Its name captures the central cross-science interpretation: markets are represented as moving through a lower-dimensional state space containing relatively persistent basins, while transitions correspond to movement between those basins and uncertainty rises near boundaries.

Six recurring motifs were especially important. The first is the free-energy or potential landscape with metastable basins separated by barriers, connecting statistical physics, chemistry, and geophysics. The second is critical slowing down, in which systems may become less resilient before a transition. The third is spectral or operator decomposition

of coupled systems, connecting signal processing, network science, and dynamical systems. The fourth is self-exciting or triggered cascades, linking nucleation and aftershock processes. The fifth is information-theoretic divergence between local distributions as a model-free change signal. The sixth is low-dimensional order-parameter reduction of high-dimensional observations.

The synthesizer also preserves ideas not selected for the first executable prototype. These include transfer-entropy network reconfiguration, topological data analysis through persistent homology, nucleation-style supercritical growth, robust-control fragility measures, and switched dynamical systems with explicit guard boundaries. Preservation matters because a first implementation should not be confused with the entire scientific frontier discovered by the swarm.

The synthesis procedure can be viewed as a translation across levels of abstraction. At the highest level, agents offer scientific narratives: markets occupy basins, approach critical points, reorganize networks, or emit weak precursor signals. At the middle level, the synthesizer extracts functional requirements: reduce dimensionality, infer persistent states, measure transition pressure, compare alternative geometries, and retain uncertainty. At the implementation level, those requirements are assigned to executable components. This translation prevents the final architecture from being a collage of metaphors. Every retained idea must perform a defined role in the computational system.

Recurrence is evaluated structurally rather than by exact vocabulary. “Order parameter,” “reaction coordinate,” “dominant mode,” and “latent factor” can all point toward a reduced coordinate that summarizes collective behavior. “Barrier crossing,” “bifurcation,” “rupture,” and “change point” can all describe a transition, but they imply different mechanisms and diagnostics. The synthesizer therefore groups proposals when they solve a common functional problem while retaining distinctions that could matter empirically. This allows convergence to be recognized without pretending that the sciences are interchangeable.

Complementarity is the second major rule. A probabilistic mixture may be useful for soft state membership but weak at describing nonlinear neighborhood geometry. Spectral clustering may reveal nonconvex structure but be difficult to estimate over the full history. A state-space velocity can provide a model-independent transition measure even when the cluster label remains unchanged. Model disagreement can represent epistemic uncertainty that no single classifier exposes. MBTA combines these elements because the regime problem contains several inferential tasks, not because an ensemble is automatically superior.

Parsimony places a necessary limit on synthesis. With fifty proposals, it would be easy to build an architecture containing dozens of indicators, kernels, graphs, thresholds, and alarms. Such a system might appear comprehensive while becoming impossible to interpret or falsify. The first MBTA prototype therefore selects a small number of conventional computational components that can instantiate the swarm’s recurring ideas. This choice sacrifices some scientific richness, but it makes the architecture reproducible and exposes which benefits come from organization rather than from algorithmic novelty.

Minority preservation addresses the opposite risk. Parsimony can erase precisely the unusual ideas that motivated a cross-science search. The notebook therefore separates “not selected now” from “rejected.” Transfer entropy, topology, nucleation, robust control, and switched dynamics remain recorded as candidate extensions with identifiable roles. Future evidence can promote one of these mechanisms if the first prototype fails. This creates an option set for revision and prevents the initial implementation from becoming an irreversible commitment.

The architecture can be summarized as four layers. The representation layer compresses correlated observables into a smaller coordinate system. The basin layer estimates categorical or probabilistic states under several geometries. The transition layer measures movement and instability independently of the categorical label. The uncertainty layer records confidence and disagreement. Each layer has a different failure mode. Representation can omit important variation; basin estimators can impose unsuitable shapes; transition metrics can respond to noise; and disagreement can arise from poor models rather than genuine ambiguity. Keeping the layers separate makes those failures diagnosable.

Synthesis also produces a provenance graph. Each architectural component can be traced backward to scientific motifs and forward to empirical diagnostics. PCA connects to order-parameter reduction and dominant modes. Mixture and clustering methods connect to metastable basins and coherent regions. Transition intensity connects to state-space velocity, precursor activity, and barrier approach. Disagreement connects to observability limits and competing representations. This traceability is central to the paper’s claim: the value of the swarm lies not only in generating an idea but in documenting how diverse conjectures were compressed into a testable specification.

WHY MBTA IS NOVEL IN THIS EXPERIMENT

The novelty does not reside in claiming that PCA, Gaussian mixtures, K-means, or spectral clustering are individually new. They are not. The novelty resides in the **scientific specification**: a swarm-generated architecture that separates representation, basin inference, transition pressure, and epistemic uncertainty, and interprets those layers through cross-disciplinary mechanisms of metastability, order-parameter reduction, barrier crossing, and model disagreement. The contribution is architectural and methodological rather than an assertion of a new clustering algorithm.

5. Executable Specification of MBTA

Notebook mapping: Cell 7 reduces the feature space, estimates competing clusterings, aligns their labels, and constructs the consensus regime series.

5.1 Representation: approximate order parameters

The first empirical layer applies Principal Component Analysis to the robustly standardized feature matrix. If $X \in \mathbb{R}^{T \times p}$ is the market-state matrix, PCA constructs orthogonal directions w_k solving

$$\max_{\|w_k\|=1} \text{Var}(Xw_k),$$

subject to orthogonality constraints. The transformed coordinates $Z = XW$ are interpreted as a compact representation of dominant market-state variation.

The cross-science interpretation is useful but should remain disciplined. PCA does not literally estimate a thermodynamic order parameter. Rather, it provides an empirical analogue: a small set of coordinates through which a high-dimensional system can be summarized. The notebook retains up to six components for diagnostics and uses a compact subset for clustering.

5.2 Parallel state estimators

MBTA does not rely on one clustering geometry. Gaussian Mixture Models provide probabilistic basin membership:

$$p(z_t = k \mid Z_t) \propto \pi_k \mathcal{N}(Z_t \mid \mu_k, \Sigma_k).$$

This is attractive for the metastable-basin interpretation because observations can have soft membership and posterior confidence.

K-means supplies a simpler Euclidean partition by minimizing within-cluster squared distance,

$$\sum_{k=1}^K \sum_{t: z_t=k} \|Z_t - \mu_k\|^2.$$

Its value is partly diagnostic: if a sophisticated partition is meaningful only under one highly specific estimator, confidence should be lower.

Spectral clustering contributes a neighborhood-based view. Instead of assuming convex clusters in Euclidean space, it constructs an affinity graph and uses eigenvectors of the graph Laplacian to reveal coherent structures. For computational tractability, the notebook applies spectral clustering to a recent subsample while GMM and K-means cover the full history.

Because unsupervised labels are arbitrary, the notebook aligns K-means and spectral labels to the GMM reference using the Hungarian assignment algorithm. This prevents a meaningless label permutation from being mistaken for disagreement.

5.3 Consensus and epistemic uncertainty

For each observation, the available aligned model labels vote on the regime. The consensus state is the modal label. More importantly, disagreement is retained:

$$D_t = 1 - \frac{\max_k n_{tk}}{M_t},$$

where n_{tk} is the number of models assigning observation t to state k and M_t is the number of available models. $D_t = 0$ indicates unanimity; larger values indicate ambiguity.

This is a significant conceptual departure from standard ensemble practice. Disagreement is not merely an inconvenience to be averaged away. In a latent-state problem it may itself be informative. Near a transition boundary, different geometric views of the same observation may reasonably disagree. MBTA therefore separates the best current state estimate from confidence in that estimate.

Several implementation decisions are required before the architecture becomes reproducible. The number of retained principal components must be declared, as must the subset used by each clustering method. Random seeds, covariance parameterization, initialization counts, convergence tolerances, and sample windows affect the resulting labels. Spectral clustering additionally requires an affinity definition and neighborhood scale. These are not minor coding details: each choice expresses assumptions about the geometry and smoothness of the latent state space. The notebook fixes them so that validation measures a defined model rather than a moving collection of defaults.

PCA serves both compression and diagnostics. Explained-variance ratios show how much of the standardized system

is summarized by the retained coordinates, while loading patterns indicate which observables contribute to each direction. A component dominated by volatility, drawdown, correlation, and VIX may be interpreted as a broad stress coordinate; another combining trend, breadth, and rotation may represent risk appetite. Such interpretations remain empirical and can change across samples. The components are therefore treated as approximate order parameters, not stable economic factors guaranteed to retain the same meaning indefinitely.

The Gaussian mixture supplies posterior membership probabilities as well as hard labels. The maximum posterior can indicate whether an observation lies deeply inside one fitted basin or near overlapping components. Yet a high posterior is conditional on the mixture model being correct. If all candidate components are misspecified, apparent confidence may be misleading. This is one reason MBTA supplements within-model probability with disagreement across models. The two measures answer different questions: posterior confidence concerns uncertainty inside one probabilistic geometry, whereas disagreement concerns sensitivity to alternative geometries.

K-means is intentionally simple. Its spherical, distance-based partitions create a useful contrast with the covariance-aware GMM. Agreement between them is informative because it shows that the state structure is not entirely dependent on flexible covariance estimation. Disagreement is also informative because it identifies observations whose assignment changes when shape assumptions change. Spectral clustering adds a third perspective based on local connectivity. Because its computational cost and out-of-sample extension are less convenient, the prototype uses it selectively rather than pretending that every model has identical coverage.

Label alignment is indispensable. Cluster identifiers have no inherent semantic order: one method's state 0 may correspond to another method's state 3. The Hungarian assignment constructs a one-to-one mapping that maximizes overlap with the reference partition. Without this step, consensus and disagreement would be dominated by arbitrary permutations. Alignment does not force the partitions to agree; it merely removes a meaningless source of disagreement so that remaining differences reflect assignments rather than names.

Consensus is calculated observation by observation from the available aligned labels. A modal vote provides a transparent operational state, but ties and missing model coverage must be governed explicitly. The associated disagreement statistic records the fraction of votes not supporting the mode. With three available models, unanimity yields zero disagreement, while a two-to-one split yields one third. The discreteness of this measure is a limitation, but its interpretation is immediate. A future implementation could add probability-weighted voting or stacking while retaining the principle that the final state and its uncertainty are separate outputs.

The executable specification is therefore more than a clustering pipeline. It produces a coordinated set of objects: reduced coordinates, model-specific labels, posterior probabilities where available, aligned labels, consensus states, and disagreement. These objects allow researchers to ask whether a result is driven by the representation, one estimator, or the consensus rule. They also support later monitoring. A production observatory can track changes in loadings, centroids, covariances, label agreement, and assignment confidence rather than waiting for realized portfolio performance to reveal that the regime model has drifted.

Reproducibility also requires preserving intermediate artifacts. The fitted scaler, PCA loadings, clustering parameters, alignment maps, and consensus rules should be serialized with version identifiers and sample dates. A regime label cannot be audited from the final chart alone. Saving the complete estimation state makes it possible to reconstruct why a particular observation received a label, compare successive model versions, and determine whether a historical revision came from new data or a changed specification.

6. Transition Dynamics: Where the Market Is Versus How It Is Moving

Notebook mapping: Cell 8 spans Sections 6 and 7, combining regime characterization, dwell and transition statistics, transition pressure, and the market-regime observatory.

State and transition are different inferential objects. A market can remain inside the same basin while moving rapidly toward its boundary. Conversely, a classification boundary can occasionally be crossed without a large economic event. The notebook therefore constructs transition intensity independently of the labels.

Let X_t denote the standardized feature vector. A simple state-space velocity is

$$v_t = \sqrt{\frac{1}{p} \sum_{j=1}^p (X_{tj} - X_{t-1,j})^2}.$$

A rolling average of v_t becomes the transition-intensity measure, and a rolling median-and-standard-deviation normalization creates a transition-pressure z-score. This is deliberately simpler than the full scientific motifs proposed by the swarm. It serves as an executable proxy for movement through state space, stress accumulation, precursor activity, and barrier approach.

Cell 8 also estimates empirical transition probabilities and dwell times. If N_{ij} is the number of observed transitions

from state i to state j , the empirical transition matrix is

$$\hat{P}_{ij} = \frac{N_{ij}}{\sum_j N_{ij}}.$$

The diagonal terms measure one-step persistence. Dwell-time analysis asks how long contiguous episodes remain in a state. These quantities connect MBTA to the language of kinetic-state models and Markov processes without requiring the consensus classifier itself to be Markovian.

Transition analysis begins by distinguishing a state change from a large market move. A label can change because an observation crosses a fitted decision boundary even when the underlying movement is modest. Conversely, a market can experience a large displacement while remaining inside a broad stress basin. The first event concerns classification geometry; the second concerns motion through the observable system. MBTA therefore records both the discrete switch indicator and a continuous transition-intensity measure. Their relationship is an empirical question rather than an identity imposed by construction.

The state-space velocity uses standardized features so that no single variable dominates merely because of its units. Taking the root mean square across feature changes produces a scalar measure of multivariate movement. This measure is deliberately symmetric: it responds to rapid improvement as well as deterioration. Direction is recovered by inspecting the underlying feature changes or the destination state. The advantage is that transition intensity remains conceptually separate from a risk score. It measures how quickly the system is moving, not whether the destination is economically favorable.

Smoothing the velocity over a rolling window reduces the influence of isolated daily noise, but it introduces a horizon choice. A short window responds quickly and may create false alarms; a long window is more stable but may identify a transition only after it is well underway. The notebook uses a transparent rolling construction because the objective is to test the layered architecture. A production system would compare several horizons, declare the decision latency associated with each, and evaluate whether alerts arrive early enough to change an actual risk process.

Normalization converts transition intensity into a local pressure statistic. Comparing the current value with a trailing median and dispersion estimate asks whether movement is unusual relative to the recent environment. This is preferable to one fixed threshold when market activity changes over time. However, local normalization can also make a sustained high-volatility period appear ordinary once the rolling baseline adapts. Absolute intensity and normalized pressure should therefore be retained together. One measures the magnitude of movement; the other measures surprise relative to recent history.

The transition matrix provides a complementary, discrete description. Its diagonal entries estimate one-step persistence, while off-diagonal entries show which state changes occur directly in the fitted history. A nearly empty transition can mean that two regimes are separated by an intermediate state, that the sample is too small, or that the clustering geometry makes direct movement unlikely. It should not automatically be interpreted as an economic impossibility. Confidence intervals or Bayesian smoothing would be valuable when rare regimes produce small transition counts.

Dwell times reveal information hidden by one-step persistence. Two states can have similar diagonal probabilities but different episode structures if one has many short visits and a few extremely long episodes. Reporting the mean, median, maximum, and full distribution of contiguous runs helps identify this asymmetry. Hazard-based analysis could go further by asking whether the probability of exit rises or falls with time already spent in a state. Such duration dependence would challenge a simple memoryless Markov interpretation and could motivate semi-Markov models.

Transitions also have direction. Entry into a stress state may occur rapidly through volatility, correlation, and drawdown, whereas exit may proceed gradually through breadth and trend recovery. A symmetric velocity statistic cannot capture this asymmetry by itself. Conditional feature trajectories around switches can show which observables typically lead, coincide with, or lag each transition type. The present notebook does not claim causal ordering, but it creates the state and transition objects needed for such event studies.

Model disagreement adds another transition lens. If alternative estimators disagree more often near switches, ambiguity may be part of the boundary phenomenon. Yet disagreement can also increase because one model becomes unstable or because spectral labels are unavailable outside their estimation window. The system must separate structural disagreement from missingness and implementation artifacts. A useful diagnostic is to compare disagreement before, during, and after transitions while also examining model-specific confidence and coverage.

These distinctions matter operationally. A risk manager may care about current state, speed of movement, pressure relative to recent history, probability of exit, likely destination, and confidence in the label. A single regime color cannot answer all of these questions. MBTA's transition layer is designed to keep them visible as separate quantities. The resulting observatory is more complex than a categorical model, but it is also more honest about the difference between location, motion, and uncertainty.

This separation also creates clearer evaluation questions. State quality can be judged through persistence and profile differentiation; transition quality can be judged through timing and association with realized disturbances; uncertainty quality can be judged through calibration against instability and model error. If one layer fails, the others need not be discarded automatically. Modular evaluation is therefore not only an engineering convenience but

a way to make the scientific claims narrower, more precise, and easier to revise.

The transition record should additionally preserve event context. For each switch or pressure spike, researchers can store the originating state, destination state, feature changes, disagreement, and subsequent path. A catalog of such events allows repeated mechanisms and false alarms to be studied without selecting examples retrospectively. Over time, this event library could support transition-specific models while retaining the transparent baseline described here.

7. Empirical Results: Four Latent Market States

The executed notebook contains 3,641 observations and thirteen standardized features. The synthesis initializes a four-state architecture, and the empirical ensemble produces four persistent consensus regimes. Their frequencies are not equal: regime 1 accounts for approximately 46.4% of observations, regimes 0 and 2 for 24.1% and 22.8%, and regime 3 for 6.7%.

Table 1: Empirical regime summary

Regime	Observations	Share	Persistence	Mean dwell	Max dwell
0	877	0.241	0.846	6.50	47
1	1689	0.464	0.909	10.97	284
2	830	0.228	0.814	5.39	41
3	245	0.067	0.922	12.89	64

The profiles show why anonymous labels should be interpreted only after estimation. Regime 1 is comparatively calm and constructive: median sixty-three-day equity trend is positive, volatility is low, drawdown is shallow, breadth is strong, and VIX is comparatively contained. Regime 3 is a stress configuration: median trend is negative, realized and downside volatility are highest, drawdown is deepest, average correlation is highest, breadth is weakest, and VIX is highest. Regime 0 exhibits weaker return/trend conditions with elevated volatility and defensive gold/bond rotation. Regime 2 combines positive short-horizon return and trend with intermediate volatility and strong breadth, but its rotation measures differ from regime 1. These descriptions are empirical interpretations, not labels imposed ex ante.

The transition matrix reinforces the regime interpretation. Diagonal probabilities range from 0.814 to 0.922. Regime 3, although rare, has the highest one-step persistence at 0.922. Regime 1 is also persistent at 0.909. Overall consensus persistence is 0.873. This is a crucial result: a classifier that redraws states independently each day may be a clusterer, but it is difficult to call it a regime model. MBTA produces substantial temporal continuity.

Table 2: Empirical one-step transition matrix

From / To	0	1	2	3
0	0.846	0.057	0.078	0.019
1	0.041	0.909	0.049	0.000
2	0.059	0.124	0.814	0.002
3	0.065	0.000	0.012	0.922

The four regimes are intentionally left with numeric identifiers in the empirical tables. Descriptive names can help communication, but they can also imply a stability and causal meaning that the estimation has not established. A label such as “crisis” may be appropriate for one historical cluster and misleading when the same centroid is encountered in a different institutional environment. The disciplined sequence is to estimate anonymous states, summarize their feature distributions, examine representative episodes, and only then assign provisional economic descriptions.

Regime 1’s large share and high persistence make it the closest analogue to a prevailing background condition. Its comparatively low volatility, shallow drawdown, positive trend, and broad participation suggest a constructive environment. Nevertheless, a frequent state is not necessarily a neutral benchmark. The sample period, asset universe, and scaling method influence which configuration occupies the largest region. In a longer history including different inflation and monetary regimes, the dominant basin could have a different profile.

Regime 3 illustrates the statistical difficulty of rare but important states. It contains only 6.7 percent of observations, yet it is highly persistent once entered and exhibits the strongest stress characteristics. Rare-state estimation is vulnerable to small-sample covariance error and sensitivity to initialization. At the same time, merging the state into a broader volatile regime could erase economically meaningful extremes. The appropriate response is not automatically to keep or discard it, but to test whether its profile, persistence, and occurrence remain stable under resampling and

alternative estimation windows.

Regimes 0 and 2 demonstrate why a multidimensional system is useful. They cannot be summarized adequately by a simple calm-versus-stress axis. Similarities in one return or volatility measure coexist with differences in trend, breadth, correlation, or defensive rotation. These distinctions may matter to portfolios whose exposures respond differently to equity direction, cross-asset dependence, and the behavior of defensive instruments. A regime architecture earns its complexity only when such multivariate differences are repeatable and decision relevant.

The transition matrix indicates that all four states have substantial one-day persistence, but persistence alone does not prove that the states are economically real. Rolling features mechanically create serial dependence because adjacent observations share much of the same input window. Cluster assignments can therefore remain stable even if the latent interpretation is weak. This is why the paper pairs persistence with feature profiles, perturbation tests, model agreement, transition diagnostics, and forward differentiation. No single metric is allowed to carry the full claim.

The observed transition paths can also reveal whether the four-state representation behaves like a line, a cycle, or a more complex graph. Frequent transitions between particular pairs suggest neighboring basins, while rare direct transitions suggest greater separation or an intermediate state. Visualizing this graph with edge weights and episode examples would help practitioners understand how the model moves rather than only how often it stays. The current matrix is the first summary of that topology.

Economic interpretation must respect the difference between conditional description and prediction. A regime's historical feature profile describes the observations used to assign the state. Forward returns and volatility describe what followed those assignments in the sample. Neither establishes that the state causes the outcome or that the relationship will persist. Overlapping windows, common macroeconomic drivers, and in-sample estimation can all inflate apparent differentiation. The results support continued testing, not automatic deployment.

The empirical outcome is therefore best read as a feasibility result. The ensemble can compress thirteen observables into four persistent, interpretable configurations without forcing all models to agree. One configuration is rare and stressed; others separate more moderate forms of market behavior. The states form a useful language for diagnostics and research. Whether they form a durable decision system depends on rolling estimation, untouched data, benchmark comparisons, and governance rules described later in the paper.

Representative dates and episodes would provide a further interpretive test. Researchers could inspect whether observations assigned to each regime correspond to recognizable combinations of market behavior without selecting only famous events. Randomly sampled episodes, nearest observations to each centroid, and low-confidence boundary cases should all be reviewed. This balanced examination would reduce hindsight storytelling and reveal whether the numeric labels remain intelligible across ordinary as well as extreme periods.

Distributional summaries should accompany those examples. Medians can hide multimodality within a state, while means can be distorted by crisis observations. Quantiles, density plots, within-state correlations, and time-varying profiles would show whether a regime is internally coherent or merely combines several distinct subconditions. If a state repeatedly decomposes into stable subgroups, that evidence would support revisiting the chosen state count or adopting a hierarchical representation.

8. Validation Philosophy: No Ground-Truth Regime Labels

Notebook mapping: Cell 9 spans Sections 8 and 9, testing empirical validity, perturbation robustness, economic differentiation, and convergence of the scientific swarm.

Unsupervised regime inference cannot be validated like a supervised classifier. There is no authoritative daily label saying "the true regime today was state 2." The notebook therefore adopts a portfolio of falsification-oriented diagnostics. Each asks a different question and none is converted into a single artificial score.

8.1 Geometric separation

The GMM silhouette score is 0.176. This is modest. It would be inappropriate to describe the four states as cleanly separated clouds. Yet this weakness is itself informative. Financial regimes overlap; transition zones exist; and the same observed configuration can plausibly belong to more than one latent interpretation. The modest silhouette score supports caution against ontological overstatement.

8.2 Temporal persistence

Consensus persistence is 0.873. This is much stronger than the geometric separation result and supports the interpretation that the inferred states possess temporal continuity. Persistence is not sufficient for usefulness—a trivial constant classifier would be perfectly persistent—but in conjunction with cross-sectional differentiation it is necessary evidence that the model is identifying conditions rather than daily noise.

8.3 Perturbation robustness

The notebook perturbs the standardized feature matrix with small Gaussian noise, rebuilds the PCA-GMM partition twenty times, and compares each perturbed partition with the baseline using Adjusted Rand Index (ARI). ARI is appropriate because it is invariant to arbitrary permutations of cluster labels.

The mean ARI is 0.870, the median is 0.949, the maximum is 0.965, and the minimum is 0.402. The central tendency is encouraging: the partition usually survives small perturbations. The minimum is an important warning: robustness is not universal. Certain perturbations materially change the partition. A rigorous interpretation must report both facts.

8.4 Transition validation and model disagreement

The experiment records 461 consensus regime switches. Mean transition intensity on switch observations is 1.054 times that on non-switch observations. The ratio is greater than one but only modestly so. The independent transition-pressure measure therefore contains some association with state changes, but the evidence is not overwhelming.

Model disagreement is more revealing. Mean disagreement across the sample is 0.144, while disagreement on switch observations relative to non-switch observations rises by a factor of 2.141. This suggests that transitions are not merely moments when the consensus label changes; they are also moments when alternative representations of the market become less unanimous. Epistemic uncertainty may therefore be a useful state variable in its own right.

8.5 Economic differentiation over a 21-day horizon

Cell 9 computes subsequent S&P 500 log return and realized volatility over twenty-one trading days conditional on the regime observed at time t . The calculation is strictly forward-looking in construction:

$$R_{t,t+21} = \log P_{t+21} - \log P_t,$$

and

$$\sigma_{t,t+21} = \text{Std}(r_{t+1}, \dots, r_{t+21})\sqrt{252}.$$

The final twenty-one observations are excluded because they lack a complete future window. This leaves 3,620 valid observations.

Table 3: Conditional 21-day forward behavior by regime

Regime	Mean return	Median return	Mean vol.	Median vol.	Approx. ann. return
0	0.0145	0.0277	0.1794	0.1624	0.174
1	0.0053	0.0121	0.1157	0.1054	0.064
2	0.0132	0.0175	0.1175	0.1042	0.158
3	0.0350	0.0475	0.2645	0.2572	0.420

The range of mean forward log return across regimes is 0.0297; the range of mean forward realized volatility is 0.1487. These differences indicate economic differentiation, especially in volatility. They do *not* establish causality or a profitable trading rule. The windows overlap, the regimes were discovered on the same historical sample, and the exercise is descriptive. The appropriate conclusion is narrower: the inferred states correspond to meaningfully different subsequent market environments, which is desirable for a latent-state representation intended for risk and allocation research.

Validation is organized as a portfolio because the diagnostics can conflict. Strong persistence may coexist with weak geometric separation. High average perturbation robustness may coexist with a damaging worst case. Forward volatility may differ clearly while forward return estimates remain noisy. Such tension should not be compressed into a single score unless the intended decision supplies defensible weights. Reporting the profile of evidence makes it possible for different users to judge the architecture according to their risk and governance needs.

The silhouette score asks whether observations are closer to their assigned cluster than to neighboring clusters under a chosen distance. A modest value is unsurprising when market conditions change continuously and boundaries overlap. It nevertheless constrains interpretation: the states should not be presented as isolated natural kinds. Additional geometry tests could include alternative distance metrics, density separation, posterior entropy, and stability of nearest neighbors. Each would illuminate a different aspect of overlap.

Perturbation robustness evaluates local sensitivity rather than truth. Adding small noise to standardized inputs approximates measurement error and minor specification variation. A high ARI indicates that most pairwise relationships survive that disturbance. It does not show that the state count is correct, that the feature universe is complete, or that a structural break will leave the partition intact. The low minimum ARI is therefore important because it identifies the existence of fragile directions even when the average result is reassuring.

Forward differentiation is evaluated after the state is observed, but a deployable test would impose stricter timing.

Features, scaling parameters, PCA loadings, mixture parameters, label mappings, and interpretation rules would all be estimated using past data only. The regime at time t would be assigned without knowledge of future outcomes, and portfolio actions would include realistic execution delays and costs. The present calculation is a descriptive bridge between latent structure and economic relevance, not a substitute for that walk-forward protocol.

Benchmarking is another necessary validation dimension. A complex architecture should be compared with simple volatility thresholds, drawdown rules, trend-volatility quadrants, HMMs, and Markov-switching models. If those alternatives provide equal stability and differentiation with lower complexity, MBTA's scientific narrative would not justify its use. Conversely, even modest improvements may be valuable if disagreement and transition pressure supply governance information absent from the benchmarks. The comparison must evaluate the complete output, not just one clustering metric.

Validation should also be conditional on market era. Results can be recomputed across monetary regimes, crisis and non-crisis periods, pre- and post-pandemic samples, or rolling windows of equal length. The purpose is not to find a partition that never changes; markets and measurement systems evolve. The purpose is to determine which elements remain stable enough to support interpretation and which require recalibration. A model that changes transparently under a declared procedure can be more useful than one that appears stable only because it is never challenged.

Uncertainty around validation metrics should be reported rather than relying only on point estimates. Block resampling can produce intervals for persistence, forward differences, and switch associations while respecting some temporal dependence. Sensitivity ranges across seeds, windows, and preprocessing choices can reveal specification risk. These intervals will often be wide, but that is useful information: a regime model should not appear more precise than the finite number of episodes and highly dependent observations permit.

9. Validating the Swarm Itself

The experiment contains two distinct objects that require validation: the empirical regime model and the scientific search process that generated its architecture. Cell 9 therefore asks whether the swarm's collective scientific signature stabilizes as more agents are included.

Prefixes of 5, 10, 20, 30, 40, and 50 proposals are compared with the full swarm. One diagnostic measures the L_1 distance between the prefix distribution of proposed state counts and the final distribution. Another measures Jaccard similarity between the prefix's most common mathematical vocabulary and the final vocabulary.

The state-count distance declines from 1.00 with five agents to 0.60 with ten, 0.20 with twenty, 0.20 with thirty, 0.11 with forty, and zero by construction at fifty. Vocabulary Jaccard similarity rises overall from 0.333 at five agents to 0.739 at forty and 1.0 at fifty, although the path is not monotonic. This is exactly the kind of pattern one would expect from a diverse scientific population: early samples are unstable; broader populations increasingly resemble the final intellectual structure; and some intermediate additions temporarily change the vocabulary rather than moving smoothly toward consensus.

This distinction is essential. **Scientific convergence among agents is not the same as empirical convergence among clustering models.** The former concerns whether the research architecture is an idiosyncratic artifact of a few prompts. The latter concerns whether different algorithms see similar structure in the market data. A serious swarm-based research process should expose both.

Swarm validation begins with coverage. The population should span the scientific families and specialties promised by its constitution, with no family silently missing because of failed calls or schema rejection. Counts alone are insufficient: the substantive fields must contain mechanisms specific to the assigned roles. A statistical-physics agent that returns a generic moving-average rule has satisfied the file format but not the scientific objective. Qualitative audits of role fidelity therefore complement automated schema checks.

The prefix experiment asks how quickly the collective signature approaches the full population. State-count distributions represent one coarse output, while mathematical vocabulary represents a broader conceptual fingerprint. Their different convergence paths are informative. State-count preferences may stabilize while the mechanisms used to justify those counts continue to evolve. Conversely, vocabulary can appear stable even when later agents change the proposed architecture. Multiple signatures are needed to assess whether the swarm has reached diminishing returns.

Ordering matters in a prefix analysis. If agents are grouped by family, early prefixes may overrepresent the first disciplines and convergence may partly reflect the order in which families are added. Randomized permutations or stratified prefixes would provide a stronger test. Repeating the analysis across many orderings could estimate the distribution of convergence curves and identify which agents contribute unusually large conceptual changes. The current deterministic prefix result is an initial diagnostic rather than a complete sampling study.

Redundancy is not automatically waste. Several agents may rediscover the same structural motif independently, increasing confidence that it is not tied to one disciplinary prompt. But repeated phrasing can also arise from a shared model prior. A useful analysis distinguishes semantic recurrence from near-duplicate language. Embedding-based clustering, manual coding, or structured motif tags could measure whether proposals converge on the same function

through genuinely different reasoning. Provenance by family would then reveal whether a motif is cross-disciplinary or concentrated in one branch.

Marginal contribution provides another view. For each agent or family, the synthesizer can be rerun without that contribution and the resulting architecture compared with the full synthesis. Large changes indicate influential proposals; negligible changes indicate redundancy under the current synthesis rule. This leave-one-out analysis would not measure scientific worth in an absolute sense, because a redundant idea may still provide independent support, but it would make architectural dependence more transparent.

The synthesizer itself is a source of model risk. A different synthesis prompt, language model, or allowed-component list might compress the same fifty proposals into another architecture. Validation should therefore include synthesis perturbations: vary the instructions, hide family labels, change proposal order, or ask independent synthesizers to produce candidate architectures under the same constraints. Agreement would support robustness; disagreement would expose choices that require human adjudication.

Self-reported novelty, plausibility, and confidence should be treated cautiously. Language models are not calibrated scientific referees, and their scores can reflect style or prompt framing. These fields are useful primarily as descriptive metadata and as a way to identify inconsistencies, such as a highly confident proposal with vague validation tests. External evaluation should rely on mechanism quality, implementability, empirical performance, and failure analysis rather than on the agents' own numerical ratings.

True swarm convergence should not mean elimination of disagreement. If every agent returns the same ontology, the population may have failed to explore the search space. The desired outcome is a stable higher-level architecture accompanied by a documented set of alternatives and unresolved questions. Convergence at the functional level can coexist with diversity at the mechanistic level. For example, several disciplines may support a transition layer while disagreeing about whether the relevant mechanism is bifurcation, nucleation, rupture, or information divergence.

Cost and latency are also part of validation. A research process that requires fifty agents should demonstrate that additional proposals produce enough conceptual coverage to justify their computational and review burden. Convergence curves can inform a stopping rule: launch new agents until state-count preferences, motif coverage, and architectural components stabilize within declared tolerances. Such a rule would turn swarm size from a round number into an empirically governed design parameter.

Finally, human review remains necessary. The swarm can widen search, standardize proposals, and reveal recurrence, but it cannot independently establish that a scientific analogy is valid in markets. Subject-matter experts must inspect whether transfer principles are coherent, whether observables actually measure the proposed constructs, and whether failure tests are meaningful. Swarm validation is therefore institutional: it evaluates the population, the synthesis procedure, the computational constraints, and the human decision process that connects them.

That review should use a documented rubric. Reviewers can score role fidelity, mechanism clarity, observable linkage, mathematical specificity, novelty relative to the corpus, implementability, and quality of falsification tests. Disagreements among reviewers should be retained rather than averaged without comment. A structured record would make it possible to compare human assessments with the synthesizer and to determine whether later empirical success was associated with identifiable proposal qualities.

10. How the Idoneous Specification Was Chosen

The phrase “choose the best model” is tempting but too strong for this experiment. There is no ground-truth regime label and no theorem establishing that MBTA dominates every possible specification. The notebook instead performs a disciplined selection of an *idoneous* architecture under the research objective.

The selection can be understood as a sequence of filters. The first filter is **scientific breadth**: fifty agents search a much larger space than a conventional finance-only model-selection exercise. The second is **recurrence**: concepts independently rediscovered across domains receive greater attention. The third is **complementarity**: rather than forcing one mechanism to solve representation, state inference, transitions, and uncertainty, the synthesis assigns different mechanisms to different layers. The fourth is **executability**: only ideas that can be connected to the available market features and implemented reproducibly enter the first empirical prototype. The fifth is **falsifiability**: the architecture must state conditions under which it should be rejected or revised.

The resulting MBTA is therefore “best” only in a conditional research sense: it is the architecture that most successfully compresses the scientific search into a parsimonious, interpretable, executable, and challengeable specification. Its components are conventional; their cross-science organization is the experiment's innovation.

The selection logic can be formalized as a sequence of gates rather than a weighted beauty contest. A proposal first has to be relevant to latent states and transitions. It then has to connect its scientific mechanism to available observables. Next, it must specify a computational operation that can be reproduced. Finally, it must generate tests capable of weakening or rejecting the claim. Ideas that fail an early gate remain in the archive but do not enter the first prototype. This avoids assigning precise numerical scores to dimensions that are difficult to commensurate.

Scientific breadth is evaluated against the population design. The selected architecture should reflect more than one

family and should not merely reproduce the language of a conventional finance prompt. MBTA meets this criterion through its landscape interpretation, low-dimensional collective coordinates, competing geometries, explicit transition motion, and preserved uncertainty. Breadth does not require one component from every discipline. It requires that the final specification capture functional insights that recur across genuinely different traditions.

Recurrence is evidence of relevance, not proof of correctness. Multiple sciences may use similar mathematical tools because those tools are broadly useful, or because the language model associates them strongly with hidden-state problems. The synthesis therefore gives recurrence greater weight when it appears through independent functional arguments and maps cleanly to empirical tests. PCA was not selected merely because many agents mentioned dimensionality reduction; it was selected because a reduced coordinate was repeatedly needed to represent collective market behavior and could be examined through loadings and stability.

Complementarity prevents the selection from collapsing into a single winner. The research problem asks for representation, state assignment, transition measurement, uncertainty, and validation. No one proposal solved every layer convincingly. The chosen architecture assigns these functions to different components while keeping the interfaces simple. This modularity also supports revision. A future nonlinear representation can replace PCA, or a semi-Markov estimator can replace the mixture, without discarding the transition and uncertainty layers.

Executability is treated as a scientific virtue. A mechanism that cannot be linked to the thirteen features or estimated reproducibly cannot be validated in this experiment. That does not make the idea false; it makes it premature for the current prototype. Preserving such proposals is important because a richer dataset may later make them executable. The boundary between selected and deferred ideas therefore depends partly on available measurement, not only on conceptual merit.

Interpretability operates at two levels. Component interpretability asks whether researchers can explain what PCA, clustering, alignment, consensus, and transition intensity do. System interpretability asks whether an observed regime can be traced through features, model assignments, confidence, and transition diagnostics. MBTA was favored because it supports both. A more powerful black-box sequence model might improve a predictive metric while making state meaning and failure analysis harder. Such a model should be compared, but performance alone would not settle the governance tradeoff.

Falsifiability provides the final gate. The architecture names conditions under which its regime count, state assignments, transition signals, or uncertainty interpretation would become untenable. This distinguishes selection from branding. The term MBTA is useful only if it organizes measurable claims; it should be abandoned if those claims fail. The explicit criteria in the next section are therefore part of the model specification, not an optional discussion appended after the results.

The choice is also conditioned on the educational objective. The paper aims to demonstrate how an AI swarm can move from scientific search to deterministic evidence. A highly specialized model with superior in-sample fit but little connection to the proposal corpus would not serve that purpose. MBTA makes the chain visible: motifs lead to layers, layers lead to code, code leads to diagnostics, and diagnostics lead to revision rules. This transparency is a selection criterion because the object being evaluated is both a regime model and a research process.

Alternative architectures remain legitimate competitors. A hidden semi-Markov model could represent duration directly. A dynamic factor model could provide time-varying latent coordinates. A Bayesian change-point system could emphasize transition timing instead of persistent clusters. A network-state model could center dependence structure. The swarm does not prove that MBTA dominates these alternatives. It explains why MBTA is an idoneous first implementation under the declared data, computational menu, and governance objectives.

Selection should ultimately be decision specific. A model intended for explanatory research may prioritize interpretability and provenance, while a real-time risk overlay may prioritize latency, calibration, and operational resilience. MBTA was selected for the former objective with an eye toward later operational testing. If the objective changes, the selection gates and their relative importance must be restated rather than assuming that one architecture is universally suitable.

The selection record should include rejected and deferred candidates alongside the chosen architecture. For each candidate, the record can state which gate it failed, what evidence would permit reconsideration, and whether the obstacle was conceptual, empirical, or operational. This prevents future researchers from repeatedly rediscovering the same rejected path and makes changes in judgment visible when new data or tools alter the feasibility of an idea.

THE SELECTION LOGIC

We did not ask fifty agents to vote for a winning algorithm. We asked them to expose the latent-state problem to fifty scientific ontologies. A separate synthesis layer then searched for recurring mechanisms and complementary functions, preserved minority ideas, and translated the result into an executable architecture. Empirical validation came *after* synthesis, not before, preventing the search process from becoming a disguised exercise in tuning to one historical score.

11. Falsification Criteria and What Would Make Us Abandon MBTA

A novel architecture becomes scientifically meaningful only when it can fail. The synthesis generated explicit falsification criteria. MBTA should be reconsidered if consensus across clustering architectures collapses out of sample; if transition-intensity events have no relationship to realized volatility, drawdown, or other transition outcomes; if critical-slowness proxies fail to precede known episodes more than chance; if the four-state count is unstable across bootstrap or rolling samples; if low-consensus periods are not associated with elevated uncertainty or instability; if simple mean/variance thresholds match or exceed the architecture; or if state centroids and covariances drift so severely out of sample that labels lose meaning.

These criteria are valuable because they convert the swarm from a creativity engine into part of a scientific workflow. The agents are not rewarded merely for novelty. They must also specify how their ideas could be wrong.

Falsification requires operational definitions. “Consensus collapses” should be translated into a monitored statistic such as declining pairwise agreement, rising disagreement, or unstable aligned labels over repeated out-of-sample windows. The relevant threshold should be chosen before examining the final test sample and should reflect the consequences of error. A research dashboard might issue a warning when agreement falls below its historical range and suspend interpretation when the decline persists. The objective is not to create one universal cutoff, but to prevent failure language from remaining rhetorical.

The state count is another testable claim. Four states emerged in this implementation, but the architecture should not protect that number from evidence. Bootstrap samples, rolling estimation windows, alternative initialization seeds, and information criteria can reveal whether four components recur or whether the solution frequently splits and merges. If two, three, or five states repeatedly provide greater stability and similar interpretability, the four-state specification loses support. A state count that survives only one full-sample fit should be treated as an artifact rather than a discovery.

State meaning must also persist enough to support longitudinal interpretation. Centroids, covariance patterns, PCA loadings, and feature rankings can be compared across training windows. Some drift is expected because markets evolve. Falsification occurs when the same numeric label changes meaning so extensively that historical comparisons become incoherent, or when alignment requires arbitrary manual decisions. Under those conditions, the model may still cluster observations, but it no longer provides a stable regime vocabulary.

The transition layer generates separate claims. High transition pressure should have an observable relationship with subsequent state changes, volatility shifts, drawdowns, correlation changes, or other declared outcomes. The benchmark is not perfect prediction. A useful measure should discriminate transition periods better than chance and better than simple alternatives such as a volatility spike or absolute-return threshold. If it produces frequent alarms without meaningful changes, or misses the transitions that matter most, its scientific interpretation as boundary pressure should be withdrawn.

Timing is central to that test. A signal that rises only after a regime switch may describe realized movement but cannot serve as an early-warning indicator. Event studies should therefore report the distribution of lead and lag times around multiple transition definitions. Researchers must declare whether the metric is intended as a contemporaneous stress gauge, an advance warning, or a confirmation tool. Failure relative to one purpose should not be hidden by silently adopting another after results are observed.

The uncertainty layer can be falsified as well. Model disagreement is useful only if high-disagreement observations are associated with lower assignment stability, greater sensitivity to perturbations, larger posterior entropy, more frequent switching, or degraded out-of-sample performance. If disagreement is driven mainly by a permanently weak estimator, then it reflects model maintenance rather than meaningful boundary ambiguity. Removing or repairing the weak model should reduce that artifact. The architecture must distinguish informative uncertainty from avoidable implementation error.

Benchmark dominance is perhaps the clearest rejection criterion. MBTA should be compared with parsimonious rules and established latent-state models under identical data and timing constraints. Comparisons should include not only average predictive metrics but also persistence, calibration, turnover, interpretability, computation, and robustness. If a simple trend-volatility quadrant or HMM matches the empirical benefits while producing fewer unstable choices, MBTA’s additional layers require strong justification. Scientific elegance cannot compensate indefinitely for inferior practical performance.

Decision tests create a higher standard. If the regimes are used for risk budgeting, scenario selection, or portfolio tilts, the complete policy must be evaluated with realistic delays, costs, constraints, and governance overrides. A statistically differentiated regime can still damage decisions through excessive turnover, unstable sizing, or false precision. Persistent underperformance relative to a fixed policy would falsify the operational use even if the descriptive model remained scientifically interesting. Model validity and decision usefulness are related but distinct claims.

The swarm process itself must be open to rejection. If repeated swarms with different prompts or models generate incompatible architectures, if role-specific agents fail to contribute domain-specific mechanisms, or if synthesis depends excessively on proposal order, the claim of disciplined cross-science search weakens. The remedy may be a revised

constitution, stronger structured fields, independent synthesis, or greater human review. If those changes do not improve stability, a smaller expert-led process may be more appropriate.

Falsification should trigger a graduated response. A warning can prompt diagnosis; persistent failure can suspend model use; structural failure can require re-estimation or architectural replacement. All responses should be documented in advance, including who has authority to declare a breach and what evidence is required for reinstatement. This governance prevents an attractive model from surviving through repeated exceptions. The purpose of explicit failure criteria is not to make abandonment easy, but to make continued use accountable to evidence.

Finally, rejection need not erase the entire research program. MBTA is modular. PCA can fail while the distinction between state, transition, and uncertainty remains valuable. The four-state partition can fail while the swarm methodology still contributes useful mechanisms. Conversely, the empirical model can perform adequately even if the cross-science narrative proves less distinctive than expected. Falsification should identify the narrowest claim contradicted by evidence and revise the architecture accordingly. This approach turns failure into structured learning rather than a binary verdict on the whole experiment.

An explicit audit calendar would keep these tests active. Some criteria can be monitored daily, such as assignment confidence and data quality; others require monthly or quarterly review, such as centroid drift, benchmark comparisons, and decision performance. Structural tests may be annual or triggered by exceptional events. Assigning frequencies and owners ensures that falsification is an operating process rather than a list consulted only after the model disappoints.

12. Limitations

Several limitations constrain interpretation. First, the experiment is observational. Conditional differences across regimes are not causal. Second, the feature universe is intentionally compact. It excludes options surfaces, macroeconomic surprises, funding markets, dealer positioning, intraday liquidity, and many other potentially important state variables. Third, PCA, GMM, K-means, and spectral clustering impose different geometries, and consensus does not eliminate model risk. Fourth, spectral clustering is estimated on a recent subsample for computational reasons, while GMM and K-means cover the full history. Fifth, overlapping twenty-one-day forward windows induce serial dependence. Sixth, the perturbation experiment measures local robustness to small feature noise, not robustness to structural breaks, alternative universes, or future market institutions.

There is also a limitation specific to the swarm. The scientific population is heterogeneous, but it is not the universe of science. The ten selected families, the five specialties within each family, the prompts, and the allowed empirical components all shape the synthesis. A different scientific constitution could produce a different architecture. This is not a defect to conceal; it is part of the provenance that makes the experiment auditable.

The first limitation is temporal leakage at the research-design level. Although the forward outcome calculation is constructed after the observation date, the feature transformations, PCA representation, cluster parameters, state count, and interpretation were developed on the full historical sample. This means the exercise describes structure in the available history rather than simulating a researcher who knew only the past. A genuine out-of-sample study must freeze every estimation and selection decision before the test interval or update them through a declared rolling procedure.

The observational design limits causal interpretation. Regimes summarize co-occurring market variables; they do not identify a structural force that causes returns or volatility. A stress regime may coincide with policy shocks, funding constraints, economic news, or investor deleveraging without distinguishing among them. Forward differences can reflect common causes, persistence in the input variables, or sample composition. The architecture is therefore suited to description, monitoring, and hypothesis generation, not causal attribution without additional identification strategies.

The asset universe creates measurement bias. Exchange-traded proxies are liquid and accessible, but they are imperfect representations of underlying markets. Bond ETFs can reflect duration, credit, and liquidity simultaneously. Commodity and currency instruments embed roll, funding, or construction choices. VIX is not a directly investable spot asset. The resulting features capture practical market manifestations rather than pure economic factors. Alternative instruments or data vendors could change the inferred geometry.

The sample length is substantial for daily analysis but limited relative to the diversity of financial institutions and policy environments. Rare crises contribute disproportionately to the stressed regime, while long calm periods dominate the common states. Historical episodes are not independent replications of one stationary process. A model that fits the available sample may fail when inflation dynamics, market structure, regulation, or central-bank behavior changes. Structural uncertainty is therefore larger than ordinary sampling error suggests.

Feature construction introduces overlapping windows and serial dependence. Trend, volatility, drawdown, and correlation measures reuse many of the same daily observations. This smoothing supports persistent states but complicates statistical inference because adjacent rows are not independent. Conventional significance tests can overstate effective sample size. Block bootstrap methods, nonoverlapping evaluation windows, or dependence-robust uncertainty estimates would be needed for formal inference.

Robust scaling reduces sensitivity to extremes but does not solve nonstationarity. Medians and interquartile ranges can drift, and full-sample normalization uses future distributional information. Rolling scaling would better mimic deployment but could change the meaning of a standardized unit over time. Expanding-window scaling preserves more continuity but adapts slowly after structural breaks. The choice affects distances, PCA loadings, and cluster assignments and should be included in sensitivity analysis.

PCA imposes linear, global structure. Important market states may lie on nonlinear manifolds or be distinguished by tail dependence rather than covariance. Components can rotate when the sample changes, complicating interpretation. Nonlinear representations could capture richer geometry but would introduce additional tuning and governance burdens. The prototype selects transparency over representational flexibility, and its conclusions are conditional on that choice.

Each clustering method has its own limitations. GMMs depend on distributional and covariance assumptions and can become unstable in small components. K-means favors Euclidean, roughly spherical partitions. Spectral clustering depends on affinity construction, can be computationally expensive, and lacks a simple out-of-sample assignment rule. Their consensus reduces dependence on one geometry but does not eliminate shared dependence on the same features and PCA representation. Ensemble agreement is not independent confirmation.

The spectral model's shorter coverage creates an asymmetry in the disagreement statistic. Observations with two available models have a different voting structure from observations with three. Comparisons through time must account for that changing denominator. A production system should either provide consistent out-of-sample spectral assignment, restrict disagreement analysis to common coverage, or report coverage-adjusted uncertainty. Otherwise apparent changes may partly reflect architecture rather than markets.

Label alignment also has limitations. The Hungarian algorithm maximizes overlap with a reference partition, but one-to-one matching can be strained when a model splits one reference state or merges two of them. Forcing a permutation may hide structural disagreement about the number or shape of states. Contingency tables and many-to-one diagnostics should accompany the aligned labels so that consensus does not obscure genuine differences in granularity.

The validation portfolio is incomplete. Perturbation tests use one noise process and a limited number of repetitions. The silhouette score reflects one geometry. Forward outcomes focus on the S&P 500 and a twenty-one-day horizon. Alternative horizons, assets, loss functions, and transition definitions may produce different conclusions. The reported metrics should be treated as an initial panel rather than a comprehensive certification.

The language-model process introduces its own reproducibility limits. Model versions, prompting behavior, structured-output capabilities, and default associations can change. Even with saved prompts, future calls may not reproduce the same proposals. Checkpointing preserves the corpus used in this experiment, but replication with another model may yield a different scientific population. This uncertainty reinforces the need to archive outputs and evaluate the resulting architecture independently of the model's authority.

Finally, interpretability is partly constructed by researchers. Feature profiles support economic descriptions, but names and narratives can encourage hindsight bias. Users may remember successful episodes and ignore counterexamples. Numeric state identifiers, representative random cases, uncertainty reporting, and explicit failure criteria mitigate this risk but cannot remove it. The strongest safeguard is to treat every interpretation as provisional and to evaluate whether it remains coherent on data not used to create it.

13. Pedagogical Interpretation for Financial Professionals

The notebook is especially useful for practitioners because it separates three activities that are often conflated in AI applications: **scientific imagination, model construction, and empirical judgment**.

The language model is strongest in the first activity. It can move rapidly across disciplinary vocabularies, recognize analogies, and formulate candidate mechanisms. But the LLM is not permitted to decide what the market data "must" show. The second activity is deterministic: Python constructs the features, scales them, performs PCA, estimates clusters, aligns labels, calculates consensus, and measures transitions. The third activity is empirical and skeptical: persistence, silhouette, ARI, forward differentiation, disagreement, and convergence are inspected without forcing them into a single score.

This architecture provides a practical governance lesson. In institutional finance, AI should not be treated as an oracle that outputs a regime label. It can instead be embedded in a research institution with distinct roles: hypothesis generation, architecture synthesis, deterministic estimation, validation, provenance, and human interpretation. The notebook therefore resembles a miniature scientific organization more than a single predictive model.

For risk managers, the distinction between state and transition is particularly useful. A stable regime label can inform the current operating environment; transition pressure can warn that the environment is moving rapidly even before classification changes; disagreement can indicate that the state estimate itself is uncertain. These three quantities answer different questions and should not be collapsed into one traffic-light signal.

For portfolio managers, the forward-behavior results suggest that regime information may be relevant to risk budgeting,

but they do not yet justify strategy deployment. A production application would require walk-forward estimation, transaction costs, decision lags, benchmark comparisons, and an untouched test sample. For senior management and boards, the important contribution is methodological: AI can expand the institution's scientific search capacity while preserving auditability and explicit falsification.

The separation of roles offers a practical blueprint for a research team. Domain specialists can define the observable system and review scientific analogies. AI agents can generate structured hypotheses under those constraints. Quantitative researchers can translate selected mechanisms into deterministic code. Independent validators can challenge stability, timing, and benchmark performance. Model-risk committees can approve intended uses and monitoring thresholds. Keeping these roles distinct reduces the risk that enthusiasm at the ideation stage determines the empirical conclusion.

For a risk manager, the observatory can be read as a set of questions rather than a trading signal. What state best describes the current configuration? How quickly is the system moving? Is that movement unusual relative to recent history? Do alternative models agree? How long has the current episode lasted, and which transitions have historically occurred from similar states? These questions support scenario discussion and escalation even when no mechanical portfolio action is attached.

For a portfolio manager, the regime label can organize conditional analysis. Expected return, volatility, correlation, drawdown, factor performance, and liquidity can be summarized by state. However, conditional averages should not be converted directly into allocations. The manager must consider estimation uncertainty, turnover, capacity, objectives, constraints, and whether the current observation resembles the historical center or a boundary case. Disagreement can be used to reduce conviction rather than to force a binary decision.

For quantitative researchers, the notebook demonstrates a disciplined distinction between exploratory and confirmatory work. The swarm and synthesis are exploratory: they generate mechanisms and select an initial architecture. The empirical notebook is partly confirmatory but still in-sample. A subsequent protocol should register benchmarks, metrics, horizons, and rejection rules before opening the test data. This sequence preserves the creative benefit of AI while limiting the temptation to reinterpret every result as support.

For model validators, provenance is as important as code review. They should be able to trace each feature to raw data, each architectural component to a stated function, each hyperparameter to a configuration, and each conclusion to a metric. They should also examine omitted alternatives and deferred minority ideas. Validation is stronger when it can ask why a component exists, not only whether the implementation matches its specification.

For senior management, the key governance question is intended use. A descriptive research dashboard, a risk-escalation input, and an automated allocation engine require different evidence and controls. The same regime model may be acceptable for discussion but unacceptable for capital decisions. Documentation should specify authorized users, update frequency, override rights, monitoring, and conditions for suspension. This prevents an educational prototype from migrating silently into production.

Communication should preserve uncertainty. Dashboards can show the consensus state alongside posterior confidence, model disagreement, transition pressure, and recent dwell time. Narrative summaries should explain which features drive the current classification and how the models differ. A single red or green indicator is visually convenient but collapses the architecture's central insight. Uncertainty should be visible enough that decision makers cannot mistake a weak consensus for a fact.

The swarm method also has pedagogical value beyond this specific model. It teaches researchers to separate analogy from implementation. A scientific metaphor becomes useful only when it identifies an object, an observable proxy, a mathematical operation, and a failure test. Students and practitioners can compare how different disciplines answer the same latent-state problem and then observe how empirical constraints eliminate or postpone attractive ideas. This is a more realistic picture of research than presenting one polished algorithm without its discarded alternatives.

The notebook can support collaborative training. Teams can assign reviewers to different scientific families, ask them to challenge transfer principles, and compare their assessments with the synthesizer. They can rerun the architecture with one family omitted or with an expanded feature set. Such exercises reveal how assumptions propagate through a research pipeline. The goal is not to teach that fifty agents are always necessary, but to demonstrate methods for organized intellectual diversity.

Several misuses should be stated explicitly. The state labels should not be interpreted as causal macroeconomic regimes without additional evidence. Transition pressure should not be marketed as a crash predictor. Historical forward returns should not be treated as guaranteed alpha. Model agreement should not be confused with truth, because the models share data and representation. Most importantly, the polished visual outputs should not substitute for access to the underlying definitions, code, and diagnostics.

The broader professional lesson is that AI can be placed inside a controlled research process. Its comparative advantage is rapid exploration across vocabularies and mechanisms. Deterministic computation supplies reproducibility. Statistical validation supplies resistance to attractive stories. Human governance supplies accountability for objectives and consequences. The value comes from this division of labor, not from assigning final authority to any one layer.

14. Research Frontier: From Static Regimes to an Adaptive Market Observatory

The natural next step is out-of-sample inference. The architecture should be estimated on a training period, calibrated on a validation period, and frozen before evaluation on an untouched test period. It should be compared directly with Hidden Markov Models, Markov-switching models, Bayesian change-point systems, simple volatility thresholds, and other conventional benchmarks.

A second extension is hierarchical latent structure. Markets plausibly contain slow macro states, medium-horizon risk regimes, and fast liquidity events simultaneously. A single categorical label forces these processes into one dimension. Hierarchical or factorial latent-state models would allow several hidden processes to coexist.

A third extension is richer transition science. The swarm's preserved minority ideas provide a roadmap: transfer-entropy networks could measure directional reorganization; persistent homology could detect changes in the shape of market dependence; nucleation-style models could distinguish false starts from propagating transitions; robust-control measures could quantify fragility; and explicit switched dynamical systems could formalize boundary crossing.

The deepest extension is recursive scientific adaptation. If the observatory experiences persistent model disagreement, falling separation, unstable state counts, or deteriorating out-of-sample behavior, those failures could trigger a new scientific swarm. The system would then not merely update parameters inside a fixed regime vocabulary; it would reopen the question of what a regime should mean. In that form, the research architecture begins to approach an adaptive scientific institution.

The first priority is a true walk-forward experiment. Training, validation, and test periods should be separated chronologically. All preprocessing, state-count choices, model parameters, label interpretation, and benchmark rules should be determined without access to the final test interval. At each update date, the system would estimate only from available history, assign the current state, and record transition and uncertainty measures. This design would reveal whether the architecture can maintain coherent labels as new observations arrive.

Online assignment requires additional engineering. PCA loadings and mixture parameters can be frozen for a period, updated on an expanding window, or re-estimated on a rolling window. Each choice balances adaptation against label continuity. When models are re-estimated, new states must be aligned not only across algorithms but also across time. Versioned centroids, transition matrices, and interpretation reports would be needed so that users know whether a label change reflects the market or a model update.

Hierarchical structure is a promising extension because financial processes operate at different horizons. A slow layer could represent inflation, growth, and policy conditions; a medium layer could represent risk appetite and cross-asset correlation; a fast layer could represent liquidity stress and event-driven dislocation. Factorial or hierarchical state models could allow these processes to coexist rather than forcing one label to summarize all of them. Their additional complexity would require stronger identifiability and validation tests.

The observable system can also be enriched. Options surfaces can describe skew, term structure, and tail insurance demand. Funding spreads and repo measures can capture balance-sheet stress. Intraday liquidity and order-flow variables can reveal fast transitions. Macroeconomic surprises and survey revisions can connect market states to information arrivals. International assets can distinguish local from global regimes. Each addition should be justified by a scientific mechanism and tested for incremental value rather than accumulated indiscriminately.

Nonlinear representations deserve comparison with PCA. Autoencoders, diffusion maps, kernel methods, and manifold learning may capture curved state spaces or interactions missed by linear components. Yet they can make out-of-sample assignment, stability, and interpretation more difficult. A useful research design would compare representations while holding the downstream state, transition, and uncertainty layers constant. This would isolate whether added flexibility improves the architecture or merely changes the visualization.

The preserved minority ideas define a transition-science agenda. Transfer entropy could test directional reorganization among assets. Persistent homology could measure changes in the topology of correlation networks. Nucleation models could distinguish local disturbances that fade from those that propagate. Critical-slowing indicators could examine rising autocorrelation or variance before selected transitions. Robust-control measures could quantify how small shocks move the system under parameter uncertainty. Each mechanism needs a null model and a declared event definition.

Duration modeling is another frontier. Hidden semi-Markov models can estimate state-specific dwell distributions rather than assuming memoryless exits. Survival models can relate exit hazards to transition pressure, disagreement, and elapsed duration. Competing-risk formulations can distinguish possible destination states. These tools would connect the empirical episode structure more directly to the architecture's metastability interpretation.

Decision-aware evaluation should follow descriptive validation. Instead of asking only whether regimes differ, researchers can ask whether the information improves a defined decision under realistic constraints. Examples include volatility targeting, scenario weighting, hedge timing, liquidity buffers, or risk-limit escalation. Utility, drawdown, turnover, transaction costs, and false-alarm costs can be evaluated jointly. The chosen loss function should reflect the user's objective rather than retrofitting a favorable metric.

Adaptive swarms require governance. A monitoring breach could trigger agents to diagnose whether the problem lies in data, representation, state geometry, transition measures, or market structure. New proposals would be compared with the archived architecture under the same validation harness. Human reviewers would decide whether to recalibrate a parameter, replace a component, expand the feature set, or reopen the regime ontology. Automatic self-modification without review would undermine the provenance the framework is designed to preserve.

Cross-model and cross-institution replication would strengthen the program. Different language models could generate independent scientific populations. Separate research teams could synthesize the same proposal corpus. Alternative datasets and markets could test whether the architecture generalizes beyond the original universe. Agreement would be informative, while disagreement would reveal which conclusions depend on the model, the human institution, or the market sample.

The long-run objective is not a permanently correct classification system. It is an observatory with memory: one that stores data definitions, model versions, proposals, synthesis decisions, diagnostics, failures, and revisions. Such a system can learn not only from new prices but from the history of its own scientific claims. That institutional memory is what turns repeated modeling into cumulative research.

Open research artifacts would accelerate that cumulative process. A future release could include frozen feature matrices, proposal schemas, anonymized agent outputs, synthesis records, configuration files, and evaluation scripts. Independent researchers could then test alternative representations or synthesis rules without reconstructing the entire pipeline. Replication would expose hidden assumptions and help distinguish findings that belong to the data from those that belong to one implementation.

15. Conclusion

The central result of this research is methodological. Market-regime discovery is reframed from a narrow model-selection exercise into a cross-science latent-state inference problem. The fifty-agent swarm is valuable not because fifty opinions are automatically better than one, but because the agents are assigned genuinely different mathematical traditions. Their collective role is to enlarge the hypothesis space before empirical discipline compresses it again.

The experiment demonstrates how this can be done operationally. Fifty structured proposals are generated across ten scientific families. A separate synthesis layer searches for recurrence and complementarity rather than majority preference. The resulting Metastable Basin-Transition Architecture separates representation, state inference, transition pressure, and uncertainty. PCA provides a low-dimensional substrate; Gaussian mixtures, K-means, and spectral clustering provide competing state views; Hungarian alignment makes those views comparable; consensus produces an operational state estimate; model disagreement preserves epistemic uncertainty; and transition intensity measures movement through observable market state space independently of the label.

The empirical evidence is deliberately mixed and therefore useful. Four states emerge across 3,641 observations. Consensus persistence is strong at 0.873. Perturbation robustness is high on average, with mean ARI of 0.870, but the minimum of 0.402 warns that local stability is not universal. Geometric separation is modest at 0.176, reminding us that financial regimes overlap. Transition intensity is only 1.054 times higher on switch observations, while model disagreement is 2.141 times higher, suggesting that uncertainty may be a more sensitive marker of boundary conditions than raw state-space speed. Forward twenty-one-day return and volatility differ across regimes, especially volatility, but these differences remain descriptive rather than causal or investable evidence.

The appropriate conclusion is therefore not that the notebook has discovered the four true regimes of financial markets. It has done something more defensible: it has created a reproducible mechanism for searching a vast scientific universe, synthesizing that search into an executable model, and exposing the model to explicit failure tests. It demonstrates a division of labor in which AI expands scientific imagination while deterministic computation and empirical validation constrain it.

For financial professionals, this may be the broader lesson. The next generation of quantitative research need not consist solely of larger models trained on larger datasets. It can also consist of *organized populations of specialized intelligences* that search across scientific traditions, preserve disagreement, formulate alternative mechanisms, and submit those mechanisms to common evidence. Such systems do not eliminate uncertainty. They make uncertainty structured, observable, and governable.

The research objective was to explore different scientific domains in search of an idoneous specification for a novel regime-detection model. MBTA is the first answer produced by that process, not the final answer to the regime problem. Its real value is that it comes with a provenance trail, a set of preserved alternatives, and explicit reasons to revise or abandon it. That is the difference between an AI-generated idea and an AI-assisted scientific architecture.

The empirical findings support cautious continuation rather than declaration of victory. The states are persistent, feature profiles are interpretable, and perturbation robustness is generally high. At the same time, geometric separation is modest, one perturbation produces much lower agreement, and transition intensity is only slightly elevated on switch dates. These mixed results are scientifically valuable because they identify where the prototype is strongest and where additional work is required.

Model disagreement is one of the most promising findings. Its substantial increase around switches suggests that ambiguity across geometries may contain information about regime boundaries. This does not prove that disagreement is an early warning signal, but it justifies treating uncertainty as an output rather than hiding it inside the consensus vote. Future work should test its timing, calibration, and incremental value relative to posterior probabilities and simple volatility measures.

The four inferred states should remain provisional. They are a compact description of one dataset under one feature system and one implementation. Their numeric labels are not universal market categories, and their economic descriptions can drift. A responsible research program will test alternative state counts, rolling stability, representative episodes, and out-of-sample label continuity before using them as a durable vocabulary.

The swarm's main contribution is procedural. It creates a documented expansion of the hypothesis space, forces proposals into a common schema, preserves minority ideas, and makes synthesis a separate step. This separation protects against the common tendency to mistake fluent generation for validated discovery. The agents propose; the synthesis organizes; deterministic code estimates; diagnostics challenge; and human governance decides intended use.

MBTA also demonstrates the benefit of modularity. Representation, basin inference, transition motion, and uncertainty can be tested and replaced independently. If PCA proves too restrictive, the state and validation layers can remain. If the four-state mixture fails, transition intensity may still be useful. If disagreement reflects one weak model, the ensemble can be revised without discarding the scientific-search process. Modular failure is easier to learn from than monolithic failure.

For practitioners, the appropriate near-term use is research and risk dialogue. The observatory can organize discussions about the current configuration, the speed of change, the degree of model agreement, and historically associated outcomes. It should not yet drive automated allocation. That transition would require walk-forward evidence, benchmarks, costs, operational controls, and an explicit policy connecting estimates to decisions.

For researchers, the experiment offers a broader view of AI-assisted quantitative work. Progress need not come only from increasing model scale or fitting complexity. It can come from designing better scientific institutions: diverse proposal generation, structured comparison, traceable synthesis, executable constraints, adversarial validation, and preserved memory. AI is most useful when it increases the range of questions while the research process increases the difficulty of accepting weak answers.

The final standard is replaceability. A scientific architecture should make clear when a better representation, state model, transition mechanism, or uncertainty measure has earned its place. It should retain enough provenance to compare the replacement fairly and enough humility to abandon an attractive narrative. MBTA is valuable to the extent that it supports this cycle of conjecture, implementation, criticism, and revision.

Accordingly, the paper's conclusion is both positive and limited. A structured fifty-agent swarm can generate a coherent cross-science regime architecture, and that architecture can be implemented and challenged with conventional tools. The resulting evidence is strong enough to motivate deeper testing but not strong enough to establish a universal regime ontology or investment rule. The experiment succeeds primarily by converting scientific diversity into a transparent sequence of claims that future data can confirm, modify, or reject.

The deepest contribution of the swarm is not that it chooses one regime model for us. It creates a disciplined institution for discovering, comparing, synthesizing, testing, and eventually replacing regime models as the market and our scientific understanding evolve.

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